

Networking Basics

IP addressing, CIDR notation, subnets, DNS and routing — the foundation for Module 04

Beginner

12 Minutes

AZ-104 · Prerequisites

Why This Matters

Module 04 (Networking) is 15–20% of the exam and assumes you can read IP addresses, CIDR notation, and basic routing/DNS concepts. This doc is a refresher — if you've never worked with subnets before, read it carefully before Lab 12.

IP Addresses & CIDR Notation

An IPv4 address (e.g., `10.0.1.4`) is 32 bits, written as four 8-bit numbers (0–255) separated by dots. **CIDR notation** (`10.0.0.0/16`) describes a *range* of addresses: the `/16` says the first 16 bits are fixed (the network portion), leaving 16 bits (65,536 addresses) free for hosts.

CIDR	Addresses	Typical use in these labs
<code>/16</code>	65,536	A Virtual Network's address space (e.g., <code>10.0.0.0/16</code>)
<code>/24</code>	256	A subnet within that VNet (e.g., <code>10.0.1.0/24</code>)
<code>/29</code>	8	A small subnet (e.g., for Azure Bastion, which requires <code>/26</code> minimum)

Example: A VNet with address space `10.0.0.0/16` can contain subnets `10.0.1.0/24`, `10.0.2.0/24`, etc. — each a `/24` slice of the `/16` range. Azure reserves the **first 4 and last 1** addresses in every subnet for its own use (e.g., in `10.0.1.0/24`, addresses `.0` – `.3` and `.255` aren't assignable to VMs), so a `/24` actually gives you 251 usable addresses.

Private vs. Public IP Addresses

- **Private IPs** (e.g., `10.x.x.x` , `172.16.x.x` – `172.31.x.x` , `192.168.x.x`) are only reachable within the VNet (or peered/connected networks). Every VM gets one.
- **Public IPs** are reachable from the internet. A VM only has one if you explicitly attach a Public IP resource to its network interface.

```
Private IP ranges (RFC 1918)
├─ 10.0.0.0    - 10.255.255.255
├─ 172.16.0.0  - 172.31.255.255
└─ 192.168.0.0 - 192.168.255.255
    Reachable only inside the VNet
    (or peered/connected networks)

Public IP
└─ Reachable from the internet
    Only if explicitly attached
    to a network interface ✓
```

Subnets & VNets

A **Virtual Network (VNet)** is an isolated network in Azure, divided into **subnets**. Resources (VMs, App Service with VNet integration, etc.) are deployed into a subnet. Subnets are used to:

- Apply different **Network Security Group (NSG)** rules to different tiers (e.g., a web subnet vs. a database subnet)
- Delegate a subnet to a specific Azure service (e.g., Azure Bastion requires its own subnet named `AzureBastionSubnet`)

DNS Basics

DNS (Domain Name System) translates names (`www.contoso.com`) to IP addresses. Key record types you'll encounter in Module 04:

Record type	Purpose	Example
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A	Maps a name to an IPv4 address	www → 203.0.113.10
CNAME	Aliases one name to another name	app → www.contoso.com
MX	Specifies mail servers for a domain	@ → mail.contoso.com

Azure provides two relevant DNS services: **Azure DNS** (hosts public/private DNS zones, Lab 16) and a built-in resolver at `168.63.129.16` that every VNet uses by default for name resolution and platform communication.

Routing Basics

Azure automatically creates **system routes** so that traffic within a VNet (and to the internet) flows correctly. You can override this with **custom route tables (UDRs — User Defined Routes)** — e.g., to force traffic through a firewall appliance. Module 04/05 labs use Network Watcher's **Next hop** tool to inspect which route a packet would take.

How This Connects to Other Topics

Glossary

Unfamiliar term? Check the **AZ-104 Glossary** for quick definitions of every concept used across these docs.

Module 04 — Networking

Everything here is foundational for **VNets & Subnets**, **Network Security Groups**, **Routing Fundamentals**, and beyond.

Key Takeaways

 **Summary**

- **CIDR notation** (/16 , /24) defines the size of an address range
- **VNets** are divided into **subnets**; Azure reserves 5 addresses per subnet
- **Private IPs** stay inside the VNet; **Public IPs** are internet-reachable
- **NSGs** filter traffic per-subnet or per-NIC
- **DNS** records (A, CNAME, MX) map names to addresses/services
- **System routes** are automatic; **UDRs** let you override them

Next Steps

1. **Recap:** Read **Cloud Computing Fundamentals** if you haven't yet
2. **Continue:** Read **Identity & Access Fundamentals** (needed for Module 01)
3. **Apply: Module 04: VNets & Subnets** — put CIDR and subnetting into practice

Azure AZ-104	Module	Generated
Professional Concept Documentation	00 — Prerequisites	June 15, 2026